# Task 4: Business Insight Report

## I. Executive Summary

The dataset reveals a clear medal-conversion gap between the two nations: USA athletes converted at 50.0% while Canadian athletes converted at 37.5% across the Milano Cortina 2026 Games. Both nations sent comparable roster sizes (USA: 46 athletes, Canada: 48 athletes), so the gap is not a quantity problem. It is an efficiency problem. For both the USOPC and Canada's Own the Podium, the 2030 strategic priority should be improving medal conversion among athletes who are already competing at the elite international level, not expanding participation.

Three findings drive all recommendations. First, preseason World Cup form is the strongest observable predictor of medal success: gold medalists entered Milano Cortina with a mean of 679 World Cup points compared to 574 for non-medalists. These signals are observable months before competition, making them the most actionable planning lever available. Second, physiology matters but is sport-specific: gold medalists showed a median VO2max of 78.9 versus 65.1 for non-medalists overall, but the gap varies dramatically by sport, cross-country skiing medalists averaged 93.9 VO2max while curling medalists averaged 53.6. A single fitness benchmark misallocates resources across such a diverse sport portfolio. Third, national systems provide a baseline advantage, but individual development pathways can override country averages, as Lucas Pinheiro Braathen's alpine skiing gold for Brazil illustrates. The correct lesson from that case is about athlete development quality, not country funding.

Three recommendations follow. First, direct funding toward athletes already ranked near the top of international competition rather than spreading resources broadly. Second, set sport-specific performance benchmarks, VO2max targets, training load ranges, and competition-form thresholds, rather than applying universal fitness standards. Third, use October and November World Cup results as an early-warning system to decide where late-season coaching, equipment, and recovery investment will generate the highest medal return. For Canada specifically, closing 25% of the current medal-rate gap with the USA is projected to yield approximately 1.5 additional medals; closing 50% yields approximately 3 additional medals. These estimates assume a stable roster of 48 athletes with a similar sport mix and are directional planning figures, not causal forecasts.

# II. Insight 1 - USA

The USA's record 12 gold medals at Milano Cortina reflect three reinforcing advantages that the data makes visible. The first is sport selection efficiency: the USA fielded athletes in sports where their medal conversion rate is highest. Freestyle skiing (100%), figure skating (80%), bobsleigh (66.7%), cross-country skiing (60%), and alpine skiing (50%) all show strong conversion rates. Critically, the USA generated medals across a broader portfolio of disciplines than Canada, reducing its dependence on any single sport.

The second advantage is physiological preparation. US athletes showed a median VO2max of 65.38 and median training load of 25.2 hours per week, comparable to Canada but distributed more heavily into sports where aerobic capacity directly predicts medal outcomes. The third advantage is preseason competitive form: USA athletes in alpine skiing, cross-country, and freestyle skiing arrived with World Cup point totals consistent with top-eight international ranking, which the model identifies as the strongest numeric predictor of medal conversion after sport type itself.

To protect this advantage into 2030, the USOPC should prioritise two actions. First, protect the preparation model that is producing results in freestyle skiing, figure skating, and cross-country skiing: altitude training blocks, structured recovery calendars, and competition scheduling that peaks athletes at the right moment. Second, address the identified underperformance sports: ice hockey (28.6% conversion despite 7 athletes), biathlon (0% from 2 athletes), and curling (0% from 2 athletes). These represent the clearest opportunity to add medals without requiring new athlete recruitment.

# III. Insight 2 - Canada

Canada's challenge is not athlete quality, it is conversion. With 48 athletes against the USA's 46, Canada had a slightly larger roster but produced 18 medals against the USA's 23. The data shows Canada's athletes are not physiologically weaker: median VO2max (65.70 vs 65.38), training load (25.0 vs 25.2 hours per week), and Olympic experience (median 2.5 previous Games vs 2.0 for the USA) are all comparable or slightly in Canada's favour. The gap is concentrated in sports where Canada sends athletes but generates few medals.

The sport-level breakdown shows where the gap is structural. Canada sends 4 athletes in alpine skiing and returns 0 medals (0% conversion), 3 athletes in biathlon for 0 medals, and 9 athletes in snowboard for 1 medal (11.1%). In contrast, Canada's strongest sports perform well: curling (100%), ice hockey (66.7%), short track (60%), freestyle skiing (57.1%), and speed skating (57.1%) all convert at high rates. The problem is not that

Canada’s best sports are failing, it is that a significant share of Canada’s roster is concentrated in sports where conversion is low.

The highest-return investment for Own the Podium is not adding athletes in new sports. It is optimising the final-stage preparation of athletes already ranking in the top eight internationally in their discipline. Equipment marginal gains, recovery protocol improvements, altitude training blocks, and specialist technical coaching targeted at athletes with a modelled medal probability above 50% are projected to generate the best return within the available four-year window. Snowboard represents a specific opportunity: 9 athletes produced just 1 medal (11.1%), suggesting systemic preparation or selection issues rather than a talent shortage.

# IV. Insight 3 - Head-to-Head

The sport-by-sport data reveals where the USA-Canada rivalry is genuinely competitive and where each nation should cede ground. Speed skating is the clearest near-parity sport: Canada converted 57.1% (4 medals from 7 athletes) and the USA 50.0% (1 medal from 2 athletes). Both nations are producing medals here; additional investment in speed skating depth from either country has a realistic chance of shifting outcomes. Freestyle skiing shows a similar pattern: USA at 100% (4 from 4) and Canada at 57.1% (4 from 7). Canada is competitive but converting at a lower rate, suggesting preparation rather than talent is the limiting factor.

Canada holds a clear structural advantage in curling (100% vs 0%), short track (60% vs 0%), and ice hockey (66.7% vs 28.6%). These are not sports to cede, they are Canada’s most reliable medal sources and should be protected with sustained investment. The USA holds a structural advantage in figure skating (80% vs 0%), cross-country skiing (60% vs 0%), and bobsleigh (66.7% vs 0%). For both nations, sports where the other country leads by more than 40 percentage points represent low-return investment targets over a four-year horizon. Neither country should redirect major funding into the other’s established dominance zones.

The sports where both nations currently produce zero medals, biathlon, luge, and Nordic combined, represent a shared development gap rather than a competitive opportunity. Both Canada and the USA have athletes in these sports but neither is converting. This argues for monitoring rather than major investment in the next cycle, unless pipeline indicators such as junior World Cup results or recent VO2max improvements suggest a breakthrough is imminent. Concentrated investment in existing strength zones and near-parity sports is expected to generate higher returns than building from zero in shared underperformance disciplines.

# V. Insight 4 - 2030 Blueprint

The predictive model identifies a clear medal-likelihood profile. The athletes most likely to medal compete in sports where the model's top-weighted features are relevant (cross-country skiing, curling, biathlon, freestyle skiing, speed skating), combine strong VO2max with high but sustainable training loads, carry at least two previous Olympic appearances, and arrive at the Games having placed in the top third of international World Cup rankings during the preseason season. Country factors (CZE, USA, ITA) provide an additional baseline, but are secondary to sport type and individual preparation signals.

For 2030 talent identification, both the USOPC and Own the Podium should adopt sport-specific benchmark targets rather than universal standards. In cross-country skiing, a VO2max above 90 and training load above 35 hours per week are consistent with the medalist profile. In alpine skiing, a World Cup top-15 finish during the 2028-29 season is a stronger predictor than physiology alone. In short track and speed skating, preseason World Cup points above the dataset median of 595 are associated with meaningfully higher medal conversion. These thresholds should be used as minimum conditions for directing late-cycle investment, not as elimination criteria for athlete selection.

As a practical implementation, both nations should use a pre-Games medal probability model to identify the ten to fifteen athletes with the highest conversion likelihood and direct late-cycle support, specialist coaching, equipment, and recovery resources, to that group specifically. For Canada, closing 25% of the current medal-rate gap with the USA is projected to yield approximately 1.5 additional medals; closing 50% approximately 3 additional medals. These estimates assume a stable 48-athlete roster and similar sport mix, and are planning figures rather than precise forecasts.

# VI. Insight 5 - Age, Career Stage, and the Development Pipeline

The age-band analysis reveals a broad peak performance window rather than a single peak age. Athletes aged 25–29 showed the highest overall medal rate (35.2%), followed closely by the 35–39 band (32.8%) and the 30–34 band (30.7%). The 20–24 band produced the lowest medal rate among prime-age athletes (16.2%), suggesting that athletes in this age group are frequently competing at the Games but have not yet reached peak conversion.

This is consistent with the role of Olympic experience: gold medalists had a median of 3.0 previous Games appearances, compared to 2.0 for non-medalists.

Two outliers define the boundaries of what the data can support. Metodej Jilek (CZE, speed skating, age 20) won gold with 5 previous Olympic appearances, an unusually experienced young athlete whose Games record overrides his age. Deanna Stellato-Dudek (CAN, figure skating, age 43) made her Olympic debut and did not medal, consistent with the model's low probability estimate for that profile combination. These cases together confirm that Olympic experience and sport-specific development trajectory matter more than chronological age alone.

The investment implication for both nations is a two-track pipeline. Athletes aged 22–26 with strong junior World Cup results should receive structured development investment now, targeting their peak conversion window at the 2030 Games when they will be 26–30. Athletes currently aged 28–34 who are already world-class in their discipline should receive the heaviest late-cycle preparation investment, as this group represents the highest near-term medal conversion probability. Athletes who are world-class but older than 36 warrant individual assessment rather than systematic investment, given the declining medal rate in the 40–45 band (11.5%).

# VII. Insight 6 - The Brazil Paradox: What Braathen Actually Tells Us

Lucas Pinheiro Braathen's alpine skiing gold medal for Brazil is the most analytically important outlier in the dataset. Brazil has a Country_GDP_per_capita of $11,000, the lowest of any nation in the dataset, and a Winter_Sport_Tradition_Index of 0.04, also the lowest. By any country-level metric, Brazil should not be producing Winter Olympic gold medalists. Yet Braathen won. The model assigned him a predicted medal probability of 0.50, which is higher than his country profile alone would suggest, because his individual signals, VO2max of 80.0 and World Cup preseason points of 529, were competitive with the medalist distribution.

The correct interpretation is not that Brazil has developed a competitive winter sports program. Braathen's competitive development ran through the Norwegian system. His result reflects an elite individual pathway, not a national Brazilian investment in winter sport infrastructure. Decision-makers should not draw policy conclusions from this case that suggest low-tradition countries can rapidly develop medal programs through undifferentiated funding. The lesson is more specific: athlete-level development pathway, current competitive form, and technical preparation must all be assessed alongside country-level baseline indicators. When those individual signals are strong, a weak national context is a secondary factor, not a disqualifying one.

For the USOPC and Own the Podium, the Braathen case supports one concrete recommendation: talent identification programs should incorporate athlete development pathway data, where they trained, which coaching system developed them, and what their international junior competition record shows, alongside the tabular variables available in the dataset. An athlete trained in a high-tradition system but competing under a different national flag represents a profile the model underweights by design.

# VIII. Limitations & Ethics

The dataset does not capture several factors known to influence medal outcomes: coaching quality, injury timing, psychological readiness under pressure, travel and scheduling fatigue, and equipment specifications. The model explains a meaningful share of the variance in medal outcomes, but a substantial portion remains outside the variables available. Predictions should be treated as probabilistic guidance, not deterministic rankings.

There are also meaningful ethical risks in applying predictive models to athlete funding and selection decisions. A model trained on historical data will systematically favour athlete profiles that have historically succeeded, which tends to reinforce advantages held by well-resourced nations, established sports, and athletes who have already accessed high-quality development pathways. Athletes from smaller nations, emerging disciplines, or non-traditional backgrounds may be undervalued by the model even when their individual potential is high. Braathen is the clearest example of this in the dataset. Funding decisions driven purely by medal probability risk compounding existing inequality in access to elite sport development.

The 2030 uplift estimates are directional planning figures that assume stable roster sizes, similar sport-mix distribution, and proportional gap closure across disciplines. They will require updating as new competition data becomes available in 2027 and 2028. Predictive analytics should complement expert judgment and performance director experience, not replace them. A balanced approach, using model outputs to prioritise resources while preserving human evaluation of individual athlete trajectories, is both more effective and more equitable than model-only decision-making.